

g2(0) cases

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abstract

We consider a negatively charged InAs/GaAs quantum dot (QD) with a resident electron embedded in a symmetric optical microcavity. The optically active transition couples the ground state, in which the QD hosts a single resident electron, to the negatively charged trion X^- of energy E_t , built of an electron singlet and a heavy hole. The energy of the fundamental cavity mode $\hbar\omega_0$ coincides with that of the trion transition, $E_t = \hbar\omega_0$, and the interaction with all other modes is neglected. The system is driven by a laser pulse of carrier frequency ω_L , detuned from the transition by $\Delta = \omega_0 - \omega_L$, and is subject to a magnetic field B_x applied in the Voigt geometry, perpendicular to the growth axis. Within the rotating wave approximation the Hamiltonian of the system, which accounts for the coupling to the cavity mode, for the interaction with the driving laser field and for the interaction with the magnetic field, reads (hereafter $\hbar = 1$)

$$\begin{aligned} \hat{H} = & E_t \sum_{\sigma=\uparrow,\downarrow} \hat{d}_\sigma^\dagger \hat{d}_\sigma + \omega_0 \sum_{\lambda=\pm} \hat{b}_\lambda^\dagger \hat{b}_\lambda \\ & + g \left(\hat{d}_\uparrow^\dagger \hat{a}_\uparrow \hat{b}_+ + \hat{d}_\downarrow^\dagger \hat{a}_\downarrow \hat{b}_- \right) + g^* \left(\hat{d}_\uparrow \hat{a}_\uparrow^\dagger \hat{b}_+^\dagger + \hat{d}_\downarrow \hat{a}_\downarrow^\dagger \hat{b}_-^\dagger \right) \\ & + \Omega_R(t) \sum_{\sigma=\uparrow,\downarrow} \left(e^{-i\omega_L t} \hat{d}_\sigma^\dagger \hat{a}_\sigma + e^{i\omega_L t} \hat{d}_\sigma \hat{a}_\sigma^\dagger \right) + \hat{H}_B \end{aligned} \quad (1)$$

where the last term describes the Zeeman interaction in the transverse field

$$\hat{H}_B = \mu_B g_e B_x \left(\hat{a}_\uparrow^\dagger \hat{a}_\uparrow + \hat{a}_\downarrow^\dagger \hat{a}_\downarrow \right) + \mu_B g_t B_x \left(\hat{d}_\uparrow^\dagger \hat{d}_\uparrow + \hat{d}_\downarrow^\dagger \hat{d}_\downarrow \right) \quad (2)$$

Here $\hat{a}_\sigma^\dagger$ is the creation operator of the resident electron, where $\sigma = \uparrow, \downarrow$ denotes the spin state $|\pm 1/2\rangle$; $\hat{d}_\sigma^\dagger$ is the creation operator of the negatively charged trion, where σ denotes the heavy hole spin state $|\pm 3/2\rangle$, the two electrons being bound in a spin singlet; $\hat{b}_\pm^\dagger = (\hat{b}_H^\dagger \pm i\hat{b}_V^\dagger)/\sqrt{2}$ is the creation operator of a cavity photon with circular polarization $\sigma_\pm$, expressed through the linearly polarized modes H and V ; g is the effective QD-cavity coupling parameter; μ_B is the Bohr magneton, g_e and g_t are the in-plane g -factors of the resident electron and of the trion, respectively.

The optical selection rules fix the spin structure of the light-matter coupling: a σ_+ photon converts a resident electron $|+1/2\rangle$ into the trion with the heavy hole $|+3/2\rangle$, and a σ_- photon converts $|-1/2\rangle$ into $|-3/2\rangle$. Consequently, only the triads $(\hat{a}_\uparrow, \hat{d}_\uparrow, \hat{b}_+)$ and $(\hat{a}_\downarrow, \hat{d}_\downarrow, \hat{b}_-)$ are coupled, and the polarization of the emitted photon is rigidly tied to the spin state of the QD. The transverse field B_x , in turn, mixes the two spin branches, which is the mechanism that correlates the polarizations of the two emitted photons.

Since the pump pulse contains a large number of photons, the corresponding mode is treated as a classical field, $g\hat{b}_H \rightarrow \Omega_R(t)$, where $\Omega_R(t)$ is the instantaneous Rabi frequency and the pulse area is $\theta = 2 \int \Omega_R(t) dt$. The pump is linearly polarized, $\mathbf{e}_H = (\boldsymbol{\sigma}_+ + \boldsymbol{\sigma}_-)/\sqrt{2}$, so that both spin branches are driven with equal amplitude, in contrast to the cross-polarized excitation schemes. The transformation to the frame rotating with the laser frequency, $\hat{d}_\sigma^\dagger \rightarrow e^{i\omega_L t} \hat{d}_\sigma^\dagger$ and $\hat{b}_\lambda^\dagger \rightarrow e^{i\omega_L t} \hat{b}_\lambda^\dagger$, removes the explicit time dependence of Eq. (??) and replaces both E_t and ω_0 by the single detuning Δ .

To describe the evolution of the system coupled to its environment, the Hamiltonian dynamics has to be supplemented by the dissipative terms of the Lindblad master equation

$$\partial_t \hat{\rho} = -i [\hat{H}, \hat{\rho}] + \sum_i \gamma_i \left(\hat{V}_i \hat{\rho} \hat{V}_i^\dagger - \frac{1}{2} \{ \hat{V}_i^\dagger \hat{V}_i, \hat{\rho} \} \right) \equiv \hat{\mathcal{L}} \hat{\rho} \quad (3)$$

where $\{\cdot, \cdot\}$ is the anticommutator, and γ_i and $\hat{V}_i$ are the rate and the jump operator of the i -th channel. Three dissipation channels are retained.

The first one is the escape of photons from the cavity mode into the environment, which is the process that delivers the photons to the detector. Its jump operator and rate are

$$\hat{V}_{\text{phot}} = \hat{b}_\pm, \quad \gamma_{\text{phot}} = \Gamma \quad (4)$$

where Γ is the width of the cavity mode.

The second channel is the pure dephasing of the optical transition induced by acoustic phonons, which is responsible for the damping of the Rabi oscillations. The corresponding jump operator is the population inversion operator of the trion transition,

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$$\hat{V}_{\text{phon}} = \sum_{\sigma=\uparrow,\downarrow} \left(\hat{d}_{\sigma}^{\dagger} \hat{d}_{\sigma} - \hat{a}_{\sigma}^{\dagger} \hat{a}_{\sigma} \right), \quad \gamma_{\text{phon}} = \gamma \Omega_R^2 \quad (5)$$

The quadratic dependence of the rate on the Rabi frequency corresponds to the high-temperature limit $T \gg \Omega_R$ of the exciton–phonon interaction, in which the phonon spectral density is sampled at the dressed-state splitting.

The third channel is the dephasing of the resident electron spin, caused by the fluctuating hyperfine field of the

lattice nuclei. It is described by the spin population difference operator

$$\hat{V}_{\text{spin}} = \hat{a}_{\uparrow}^{\dagger} \hat{a}_{\uparrow} - \hat{a}_{\downarrow}^{\dagger} \hat{a}_{\downarrow} + \hat{d}_{\uparrow}^{\dagger} \hat{d}_{\uparrow} - \hat{d}_{\downarrow}^{\dagger} \hat{d}_{\downarrow}, \quad \gamma_{\text{spin}} = 1/T_2^* \quad (6)$$

This channel destroys the coherence between the two spin branches and therefore directly degrades the polarization entanglement of the emitted photon pair.